

Summary

```
Test script: xthScript
xic-build: OK
Test collection: xic (Test --help)
  []: OK
xic (Test --help): 1 out of 1 tests succeeded.
Test collection: xic (Test --irgen)
  ex01.xi: OK
  ack.xi: OK
  primes.xi: OK
xic (Test --irgen): 3 out of 3 tests succeeded.
  Number of IRs: 3
  Number of canonical IRs: 3
  Number of constant-folded IRs: 3
  Number of correct IRs: 3
Test collection: xic (Test --irgen -O)
  ex01.xi: OK
  ack.xi: OK
  primes.xi: OK
xic (Test --irgen -O): 3 out of 3 tests succeeded.
  Number of IRs: 3
  Number of canonical IRs: 3
  Number of correct IRs: 3
Test collection: xic-ref-pre (--irgen [pretest])
  test_conv.xi: OK
  test_io.xi: OK
xic-ref-pre (--irgen [pretest]): 2 out of 2 tests succeeded.
  Number of IRs: 2
  Number of canonical IRs: 2
  Number of constant-folded IRs: 2
  Number of correct IRs: 2
Test collection: xic-ref-pre (--irgen -O [pretest])
  test_conv.xi: OK
  test_io.xi: OK
xic-ref-pre (--irgen -O [pretest]): 2 out of 2 tests succeeded.
  Number of IRs: 2
  Number of canonical IRs: 2
  Number of correct IRs: 2
Test collection: xic-ref (--irgen [basic test])
  arracc01.xi: OK
  arracc02.xi: OK
  arracc03.xi: Syntax error for input symbol "NUMBER" spanning from 1:1 to 1:3
    instead expected token classes are []
    Couldn't repair and continue parse for input symbol "NUMBER" spanning from 1:1 to 1:3

    IR syntax error: Can't recover from previous error(s)
  arracc04.xi: Syntax error for input symbol "NUMBER" spanning from 1:1 to 1:3
    instead expected token classes are []
    Couldn't repair and continue parse for input symbol "NUMBER" spanning from 1:1 to 1:3

    IR syntax error: Can't recover from previous error(s)
  arracc05.xi: Mismatch detected at line 1 of file arracc05.ir.nml
    expected: _I_outOfBounds_p called
    found   : 1
  arracc06.xi: Mismatch detected at line 1 of file arracc06.ir.nml
    expected: _I_outOfBounds_p called
    found   : 6976351429814890575
  arracc07.xi: OK
  arracc08.xi: Syntax error for input symbol "NUMBER" spanning from 1:1 to 1:3
    instead expected token classes are []
```

Summary

Couldn't repair and continue parse for input symbol "NUMBER" spanning from 1:1 to 1:3

IR syntax error: Can't recover from previous error(s)

arrinit01.xi: OK

arrinit02.xi: OK

arrinit03.xi: Syntax error for input symbol "NUMBER" spanning from 1:1 to 1:3

instead expected token classes are []

Couldn't repair and continue parse for input symbol "NUMBER" spanning from 1:1 to 1:3

IR syntax error: Can't recover from previous error(s)

arrinit04.xi: Syntax error for input symbol "NUMBER" spanning from 1:1 to 1:3

instead expected token classes are []

Couldn't repair and continue parse for input symbol "NUMBER" spanning from 1:1 to 1:3

IR syntax error: Can't recover from previous error(s)

arrinit05.xi: OK

arrinit06.xi: Syntax error for input symbol "NUMBER" spanning from 1:1 to 1:3

instead expected token classes are []

Couldn't repair and continue parse for input symbol "NUMBER" spanning from 1:1 to 1:3

IR syntax error: Can't recover from previous error(s)

assign01.xi: OK

assign02.xi: OK

assign03.xi: OK

assign04.xi: OK

assign05.xi: Syntax error for input symbol "ESEQ" spanning from 14:6 to 14:10

instead expected token classes are []

Couldn't repair and continue parse for input symbol "ESEQ" spanning from 14:6 to 14:10

IR syntax error: Can't recover from previous error(s)

assign06.xi: Syntax error for input symbol "ESEQ" spanning from 14:6 to 14:10

instead expected token classes are []

Couldn't repair and continue parse for input symbol "ESEQ" spanning from 14:6 to 14:10

IR syntax error: Can't recover from previous error(s)

assign07.xi: Syntax error for input symbol "ESEQ" spanning from 17:6 to 17:10

instead expected token classes are []

Couldn't repair and continue parse for input symbol "ESEQ" spanning from 17:6 to 17:10

IR syntax error: Can't recover from previous error(s)

assign08.xi: Syntax error for input symbol "ESEQ" spanning from 17:6 to 17:10

instead expected token classes are []

Couldn't repair and continue parse for input symbol "ESEQ" spanning from 17:6 to 17:10

IR syntax error: Can't recover from previous error(s)

assign09.xi: Syntax error for input symbol "NUMBER" spanning from 1:1 to 1:3

instead expected token classes are []

Couldn't repair and continue parse for input symbol "NUMBER" spanning from 1:1 to 1:3

IR syntax error: Can't recover from previous error(s)

assign10.xi: Syntax error for input symbol "NUMBER" spanning from 1:1 to 1:3

instead expected token classes are []

Couldn't repair and continue parse for input symbol "NUMBER" spanning from 1:1 to 1:3

IR syntax error: Can't recover from previous error(s)

binary01.xi: OK

binary02.xi: OK

binary03.xi: OK

binary04.xi: OK

binary05.xi: OK

Summary

binary06.xi: OK
binary07.xi: OK
binary08.xi: OK
binary09.xi: Syntax error for input symbol "EOF" spanning from 3:1 to 3:1
 instead expected token classes are []
 Couldn't repair and continue parse for input symbol "EOF" spanning from 3:1 to 3:1

 IR syntax error: Can't recover from previous error(s)
binary10.xi: Syntax error for input symbol "EOF" spanning from 3:1 to 3:1
 instead expected token classes are []
 Couldn't repair and continue parse for input symbol "EOF" spanning from 3:1 to 3:1

 IR syntax error: Can't recover from previous error(s)
binary11.xi: OK
binary12.xi: OK
block01.xi: OK
block02.xi: OK
charlit01.xi: OK
charlit02.xi: OK
functioncall01.xi: OK
functioncall02.xi: OK
functioncall03.xi: OK
functioncall04.xi: OK
if01.xi: OK
if02.xi: OK
if03.xi: OK
if04.xi: OK
if05.xi: OK
if06.xi: OK
if07.xi: OK
if08.xi: OK
if09.xi: OK
if10.xi: OK
intl01.xi: OK
intl02.xi: OK
length01.xi: OK
length02.xi: Syntax error for input symbol "NUMBER" spanning from 1:1 to 1:3
 instead expected token classes are []
 Couldn't repair and continue parse for input symbol "NUMBER" spanning from 1:1 to 1:3

 IR syntax error: Can't recover from previous error(s)
localdecl01.xi: OK
localdecl02.xi: OK
localdecl03.xi: OK
localdecl04.xi: OK
localdecl05.xi: OK
localdecl06.xi: Syntax error for input symbol "NUMBER" spanning from 1:1 to 1:3
 instead expected token classes are []
 Couldn't repair and continue parse for input symbol "NUMBER" spanning from 1:1 to 1:3

 IR syntax error: Can't recover from previous error(s)
localdecl07.xi: OK
localdecl08.xi: Syntax error for input symbol "EOF" spanning from 3:1 to 3:1
 instead expected token classes are []
 Couldn't repair and continue parse for input symbol "EOF" spanning from 3:1 to 3:1

 IR syntax error: Can't recover from previous error(s)
localdecl09.xi: OK
localdecl10.xi: Syntax error for input symbol "EOF" spanning from 3:1 to 3:1
 instead expected token classes are []

Summary

Couldn't repair and continue parse for input symbol "EOF" spanning from 3:1 to 3:1

IR syntax error: Can't recover from previous error(s)

localdecl11.xi: Syntax error for input symbol "EOF" spanning from 3:1 to 3:1

instead expected token classes are []

Couldn't repair and continue parse for input symbol "EOF" spanning from 3:1 to 3:1

IR syntax error: Can't recover from previous error(s)

localdecl12.xi: Syntax error for input symbol "EOF" spanning from 3:1 to 3:1

instead expected token classes are []

Couldn't repair and continue parse for input symbol "EOF" spanning from 3:1 to 3:1

IR syntax error: Can't recover from previous error(s)

localdecl13.xi: Syntax error for input symbol "NUMBER" spanning from 1:1 to 1:2

instead expected token classes are []

Couldn't repair and continue parse for input symbol "NUMBER" spanning from 1:1 to 1:2

IR syntax error: Can't recover from previous error(s)

localdecl14.xi: Syntax error for input symbol "NUMBER" spanning from 1:1 to 1:2

instead expected token classes are []

Couldn't repair and continue parse for input symbol "NUMBER" spanning from 1:1 to 1:2

IR syntax error: Can't recover from previous error(s)

procedurecall01.xi: OK

procedurecall02.xi: OK

procedurecall03.xi: OK

procedurecall04.xi: OK

procedurecall05.xi: OK

procedurecall06.xi: OK

stringlit01.xi: OK

stringlit02.xi: OK

unary01.xi: OK

unary02.xi: OK

unary03.xi: OK

unary04.xi: OK

while01.xi: OK

while02.xi: OK

while03.xi: OK

while04.xi: OK

xic-ref (--irgen [basic test]): 64 out of 88 tests succeeded.

Number of IRs: 66

Number of canonical IRs: 66

Number of constant-folded IRs: 66

Number of correct IRs: 64

xthScript: 76 out of 100 tests succeeded.

xic-ref (--irgen [basic test]): arracc03.xi

Syntax error for input symbol "NUMBER" spanning from 1:1 to 1:3

instead expected token classes are []

Couldn't repair and continue parse for input symbol "NUMBER" spanning from 1:1 to 1:3

IR syntax error: Can't recover from previous error(s)

Content of test case:

pa/pa4/grading/tests/base/arracc03.xi

```
1 use io
2 use conv
3
4 main(args: int[][]) {
5   a: int[5][4]
6   {
7     i: int = 0
8     while (i < length(a)) {
9       j: int = 0
10      while (j < length(a[i])) {
11        a[i][j] = i + j
12        j = j + 1
13      }
14      i = i + 1
15    }
16  }
17  {
18    i: int = 0
19    while (i < length(a)) {
20      j: int = 0
21      while (j < length(a[i])) {
22        println(unparseInt(a[i][j]))
23        j = j + 1
24      }
25      i = i + 1
26    }
27  }
28 }
```

Compiler's standard output:

Syntax error at /home/vagrant/shared/xth-

workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arracc03.xi: 10:26: Unexpected token [

Compiler's standard error:

null

Generated result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arracc03.ir.nml

Expected result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arracc03.irsol.nml

```
1 0
2 1
3 2
4 3
5 1
6 2
7 3
8 4
9 2
10 3
```

xic-ref (--irgen [basic test]): arracc03.xi

11 4
12 5
13 3
14 4
15 5
16 6
17 4
18 5
19 6
20 7

Generated result for --irgen:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arracc03.ir

1 10:26 error:Unexpected token [
2

xic-ref (--irgen [basic test]): arracc04.xi

Syntax error for input symbol "NUMBER" spanning from 1:1 to 1:3

instead expected token classes are []

Couldn't repair and continue parse for input symbol "NUMBER" spanning from 1:1 to 1:3

IR syntax error: Can't recover from previous error(s)

Content of test case:

pa/pa4/grading/tests/base/arracc04.xi

```
1 use io
2 use conv
3
4 main(args: int[][] ) {
5   a: int[5][4]
6   {
7     i: int = 0
8     while (i < length(a)) {
9       j: int = 0
10      while (j < length(a[i])) {
11        a[i][j] = i + j
12        j = j + 1
13      }
14      i = i + 1
15    }
16  }
17  {
18    i: int = 0
19    while (i < length(a)) {
20      j: int = 0
21      while (j < length(a[i])) {
22        println(unparseInt(a[f(i)][f(j)]))
23        j = j + 1
24      }
25      i = i + 1
26    }
27  }
28 }
29
30 f(i: int): int {
31   print("Index: ")
32   println(unparseInt(i))
33   return i
34 }
```

Compiler's standard output:

Syntax error at /home/vagrant/shared/xth-

workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arracc04.xi: 10:26: Unexpected token [

Compiler's standard error:

null

Generated result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arracc04.ir.nml

Expected result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arracc04.irsol.nml

```
1 Index: 0
2 Index: 0
3 0
4 Index: 0
```

xic-ref (--irgen [basic test]): arracc04.xi

```
5 Index: 1
6 1
7 Index: 0
8 Index: 2
9 2
10 Index: 0
11 Index: 3
12 3
13 Index: 1
14 Index: 0
15 1
16 Index: 1
17 Index: 1
18 2
19 Index: 1
20 Index: 2
21 3
22 Index: 1
23 Index: 3
24 4
25 Index: 2
26 Index: 0
27 2
28 Index: 2
29 Index: 1
30 3
31 Index: 2
32 Index: 2
33 4
34 Index: 2
35 Index: 3
36 5
37 Index: 3
38 Index: 0
39 3
40 Index: 3
41 Index: 1
42 4
43 Index: 3
44 Index: 2
45 5
46 Index: 3
47 Index: 3
48 6
49 Index: 4
50 Index: 0
51 4
52 Index: 4
53 Index: 1
54 5
55 Index: 4
56 Index: 2
57 6
58 Index: 4
59 Index: 3
60 7
```

Generated result for --irgen:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arracc04.ir

1 10:26 error:Unexpected token [

xic-ref (--irgen [basic test]): arracc05.xi

Mismatch detected at line 1 of file arracc05.ir.nml
expected: `_I_outOfBounds_p` called
found : 1

Content of test case:

pa/pa4/grading/tests/base/arracc05.xi

```
1 use io
2 use conv
3
4 main(args: int[][]) {
5   a: int[1]
6   println(unparseInt(a[-1]))
7 }
```

Generated result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arracc05.ir.nml

```
1 1
```

Expected result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arracc05.irsol.nml

```
1 _I_outOfBounds_p called
```

Generated result for --irgen:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arracc05.ir

```
1 (COMPUNIT
2   arracc05
3   (FUNC
4     _Imain_pai
5     (SEQ
6       (MOVE (TEMP args_main) (TEMP _ARG0))
7       (MOVE (TEMP _LEN_0) (CONST 1))
8       (MOVE (TEMP _ARR_0)
9         (CALL (NAME _I_alloc_i) (MUL (ADD (TEMP _LEN_0) (CONST 1)) (CONST 8))))
10      (MOVE (MEM (TEMP _ARR_0)) (TEMP _LEN_0))
11      (MOVE (TEMP _ARR_0) (ADD (TEMP _ARR_0) (CONST 8)))
12      (MOVE (TEMP a_main) (TEMP _ARR_0))
13      (MOVE (TEMP __var_2)
14        (CALL (NAME _IunparseInt_ain) (MEM (ADD (TEMP a_main) (CONST -8)))))
15      (MOVE (TEMP __var_1) (CALL (NAME _Iprintln_pai) (TEMP __var_2)))
16      (RETURN)))
17
```

xic-ref (--irgen [basic test]): arracc06.xi

Mismatch detected at line 1 of file arracc06.ir.nml
expected: `_I_outOfBounds_p` called
found : 6976351429814890575

Content of test case:

pa/pa4/grading/tests/base/arracc06.xi

```
1 use io
2 use conv
3
4 main(args: int[][]) {
5   a: int[0]
6   println(unparseInt(a[0]))
7 }
```

Generated result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arracc06.ir.nml

1 6976351429814890575

Expected result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arracc06.irsol.nml

1 `_I_outOfBounds_p` called

Generated result for --irgen:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arracc06.ir

```
1 (COMPUNIT
2   arracc06
3   (FUNC
4     _Imain_paai
5     (SEQ
6       (MOVE (TEMP args_main) (TEMP _ARG0))
7       (MOVE (TEMP _LEN_0) (CONST 0))
8       (MOVE (TEMP _ARR_0)
9         (CALL (NAME _I_alloc_i) (MUL (ADD (TEMP _LEN_0) (CONST 1)) (CONST 8))))
10      (MOVE (MEM (TEMP _ARR_0)) (TEMP _LEN_0))
11      (MOVE (TEMP _ARR_0) (ADD (TEMP _ARR_0) (CONST 8)))
12      (MOVE (TEMP a_main) (TEMP _ARR_0))
13      (MOVE (TEMP __var_2)
14        (CALL (NAME _IunparseInt_aai) (MEM (ADD (TEMP a_main) (CONST 0)))))
15      (MOVE (TEMP __var_1) (CALL (NAME _Iprintln_pai) (TEMP __var_2)))
16      (RETURN)))
17
```

xic-ref (--irgen [basic test]): arracc08.xi

Syntax error for input symbol "NUMBER" spanning from 1:1 to 1:3
instead expected token classes are []
Couldn't repair and continue parse for input symbol "NUMBER" spanning from 1:1 to 1:3

IR syntax error: Can't recover from previous error(s)

Content of test case:

pa/pa4/grading/tests/base/arracc08.xi

```
1 use io
2 use conv
3
4 main(args: int[][] ) {
5   a: int[5][4]
6   {
7     i: int = 0
8     while (i < length(a)) {
9       j: int = 0
10      while (j < length(a[i])) {
11        a[i][j] = i + j
12        j = j + 1
13      }
14      i = i + 1
15    }
16  }
17  println(unparseInt(a[f(a, 4, 2)-3][f(a, 4, 2)-6]))
18 }
19
20 f(a: int[][] , i: int, j: int): int {
21   print("Index: ")
22   print(unparseInt(i))
23   print(" ")
24   println(unparseInt(j))
25   a[i][j] = a[i][j] + 1
26   return a[i][j]
27 }
```

Compiler's standard output:

Syntax error at /home/vagrant/shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arracc08.xi: 10:26: Unexpected token [

Compiler's standard error:

null

Generated result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arracc08.ir.nml

Expected result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arracc08.irsol.nml

```
1 Index: 4 2
2 Index: 4 2
3 8
```

Generated result for --irgen:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arracc08.ir

```
1 10:26 error:Unexpected token [
2
```

xic-ref (--irgen [basic test]): arrinit03.xi

Syntax error for input symbol "NUMBER" spanning from 1:1 to 1:3
instead expected token classes are []
Couldn't repair and continue parse for input symbol "NUMBER" spanning from 1:1 to 1:3

IR syntax error: Can't recover from previous error(s)

Content of test case:

pa/pa4/grading/tests/base/arrinit03.xi

```
1 use io
2 use conv
3
4 main(args: int[][]){
5   a: int[][] = {{1}, {2, 3}, {4, 5, 6}, {7, 8, 9, 10}}
6   {
7     i: int = 0
8     while (i < length(a)) {
9       j: int = 0
10      while (j < length(a[i])) {
11        println(unparseInt(a[i][j]))
12        j = j + 1
13      }
14      i = i + 1
15    }
16  }
17 }
```

Compiler's standard output:

Syntax error at /home/vagrant/shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arrinit03.xi: 10:26: Unexpected token [

Compiler's standard error:

null

Generated result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arrinit03.ir.nml

Expected result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arrinit03.irsol.nml

```
1 1
2 2
3 3
4 4
5 5
6 6
7 7
8 8
9 9
10 10
```

Generated result for --irgen:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arrinit03.ir

```
1 10:26 error:Unexpected token [
2
```

xic-ref (--irgen [basic test]): arrinit04.xi

Syntax error for input symbol "NUMBER" spanning from 1:1 to 1:3

instead expected token classes are []

Couldn't repair and continue parse for input symbol "NUMBER" spanning from 1:1 to 1:3

IR syntax error: Can't recover from previous error(s)

Content of test case:

pa/pa4/grading/tests/base/arrinit04.xi

```
1 use io
2 use conv
3
4 main(args: int[][]) {
5   a: int[][] = {{f(1)}, {f(2), f(3)}, {f(4), f(5), f(6)}, {f(7), f(8), f(9), f(10)}}
6   {
7     i: int = 0
8     while (i < length(a)) {
9       j: int = 0
10      while (j < length(a[i])) {
11        println(unparseInt(a[i][j]))
12        j = j + 1
13      }
14      i = i + 1
15    }
16  }
17 }
18
19 f(i: int): int {
20   print("f() called: ")
21   println(unparseInt(i))
22   return i
23 }
```

Compiler's standard output:

Syntax error at /home/vagrant/shared/xth-

workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arrinit04.xi: 10:26: Unexpected token [

Compiler's standard error:

null

Generated result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arrinit04.ir.nml

Expected result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arrinit04.irsol.nml

```
1 f() called: 1
2 f() called: 2
3 f() called: 3
4 f() called: 4
5 f() called: 5
6 f() called: 6
7 f() called: 7
8 f() called: 8
9 f() called: 9
10 f() called: 10
11 1
12 2
13 3
14 4
15 5
```

xic-ref (--irgen [basic test]): arrinit04.xi

16 6
17 7
18 8
19 9
20 10

Generated result for --irgen:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arrinit04.ir

1 10:26 error:Unexpected token [
2

xic-ref (--irgen [basic test]): arrinit06.xi

Syntax error for input symbol "NUMBER" spanning from 1:1 to 1:3

instead expected token classes are []

Couldn't repair and continue parse for input symbol "NUMBER" spanning from 1:1 to 1:3

IR syntax error: Can't recover from previous error(s)

Content of test case:

pa/pa4/grading/tests/base/arrinit06.xi

```
1 use io
2 use conv
3
4 main(args: int[][]) {
5   x: int[] = {1, 2, 3}
6   a: int[][] = {{f(x, 1)}, {f(x, 2), f(x, 3)}, {f(x, 4), f(x, 5), f(x, 6)},
7     {f(x, 7), f(x, 8), f(x, 9), f(x, 10)}}
8   {
9     i: int = 0
10    while (i < length(a)) {
11      j: int = 0
12      while (j < length(a[i])) {
13        println(unparseInt(a[i][j]))
14        j = j + 1
15      }
16      i = i + 1
17    }
18  }
19 }
20
21 f(x: int[], y: int): int {
22   print("f() called: ")
23   println(unparseInt(y))
24   x[0] = x[0] + 1
25   return x[0]
26 }
```

Compiler's standard output:

Syntax error at /home/vagrant/shared/xth-

workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arrinit06.xi: 12:26: Unexpected token [

Compiler's standard error:

null

Generated result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arrinit06.ir.nml

Expected result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arrinit06.irsol.nml

```
1 f() called: 1
2 f() called: 2
3 f() called: 3
4 f() called: 4
5 f() called: 5
6 f() called: 6
7 f() called: 7
8 f() called: 8
9 f() called: 9
10 f() called: 10
11 2
12 3
```

xic-ref (--irgen [basic test]): arrinit06.xi

13 4
14 5
15 6
16 7
17 8
18 9
19 10
20 11

Generated result for --irgen:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/arrinit06.ir

1 12:26 error:Unexpected token [
2

xic-ref (--irgen [basic test]): assign05.xi

Syntax error for input symbol "ESEQ" spanning from 14:6 to 14:10

instead expected token classes are []

Couldn't repair and continue parse for input symbol "ESEQ" spanning from 14:6 to 14:10

IR syntax error: Can't recover from previous error(s)

Content of test case:

pa/pa4/grading/tests/base/assign05.xi

```
1 use io
2 use conv
3
4 main(args: int[][] ) {
5   a: int[3]
6   a[f(0)] = 1
7   a[f(1)] = 2
8   a[f(2)] = 3
9   println(unparseInt(length(a)))
10  println(unparseInt(a[1]))
11 }
12
13 f(x: int): int {
14   print("f() called: ")
15   println(unparseInt(x))
16   return x
17 }
```

Generated result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/assign05.ir.nml

Expected result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/assign05.irsol.nml

```
1 f() called: 0
2 f() called: 1
3 f() called: 2
4 3
5 2
```

Generated result for --irgen:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/assign05.ir

```
1 (COMPUNIT
2   assign05
3   (FUNC
4     _Imain_pai
5     (SEQ
6       (MOVE (TEMP args_main) (TEMP _ARG0))
7       (MOVE (TEMP _LEN_0) (CONST 3))
8       (MOVE (TEMP _ARR_0)
9         (CALL (NAME _I_alloc_i) (MUL (ADD (TEMP _LEN_0) (CONST 1)) (CONST 8))))
10      (MOVE (MEM (TEMP _ARR_0)) (TEMP _LEN_0))
11      (MOVE (TEMP _ARR_0) (ADD (TEMP _ARR_0) (CONST 8)))
12      (MOVE (TEMP a_main) (TEMP _ARR_0))
13      (MOVE
14        (ESEQ
15          (SEQ
16            (MOVE (TEMP _var_11) (TEMP a_main))
17            (MOVE (TEMP __var_1) (CALL (NAME _If_ii) (CONST 0))))
18            (MEM (ADD (TEMP _var_11) (MUL (CONST 8) (TEMP __var_1))))
19            (CONST 1))
20      (MOVE
```

```

21      (ESEQ
22      (SEQ
23      (MOVE (TEMP _var_12) (TEMP a_main))
24      (MOVE (TEMP __var_2) (CALL (NAME _If_ii) (CONST 1))))
25      (MEM (ADD (TEMP _var_12) (MUL (CONST 8) (TEMP __var_2)))))
26      (CONST 2))
27      (MOVE
28      (ESEQ
29      (SEQ
30      (MOVE (TEMP _var_13) (TEMP a_main))
31      (MOVE (TEMP __var_3) (CALL (NAME _If_ii) (CONST 2))))
32      (MEM (ADD (TEMP _var_13) (MUL (CONST 8) (TEMP __var_3)))))
33      (CONST 3))
34      (MOVE (TEMP __var_5)
35      (CALL (NAME _IunparseInt_aii) (MEM (SUB (TEMP a_main) (CONST 8)))))
36      (MOVE (TEMP __var_4) (CALL (NAME _Iprintln_pai) (TEMP __var_5)))
37      (MOVE (TEMP __var_7)
38      (CALL (NAME _IunparseInt_aii) (MEM (ADD (TEMP a_main) (CONST 8)))))
39      (MOVE (TEMP __var_6) (CALL (NAME _Iprintln_pai) (TEMP __var_7)))
40      (RETURN)))
41      (FUNC
42      _If_ii
43      (SEQ
44      (MOVE (TEMP x_f) (TEMP _ARG0))
45      (MOVE (TEMP _ARR_1) (CALL (NAME _I_alloc_i) (CONST 104)))
46      (MOVE (MEM (TEMP _ARR_1)) (CONST 12))
47      (MOVE (TEMP _ARR_1) (ADD (TEMP _ARR_1) (CONST 8)))
48      (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 0))) (CONST 102))
49      (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 8))) (CONST 40))
50      (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 16))) (CONST 41))
51      (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 24))) (CONST 32))
52      (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 32))) (CONST 99))
53      (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 40))) (CONST 97))
54      (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 48))) (CONST 108))
55      (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 56))) (CONST 108))
56      (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 64))) (CONST 101))
57      (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 72))) (CONST 100))
58      (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 80))) (CONST 58))
59      (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 88))) (CONST 32))
60      (MOVE (TEMP __var_8) (CALL (NAME _Iprint_pai) (TEMP _ARR_1)))
61      (MOVE (TEMP __var_10) (CALL (NAME _IunparseInt_aii) (TEMP x_f)))
62      (MOVE (TEMP __var_9) (CALL (NAME _Iprintln_pai) (TEMP __var_10)))
63      (MOVE (TEMP _RET0) (TEMP x_f))
64      (RETURN)
65      (RETURN)))
66

```

xic-ref (--irgen [basic test]): assign06.xi

Syntax error for input symbol "ESEQ" spanning from 14:6 to 14:10

instead expected token classes are []

Couldn't repair and continue parse for input symbol "ESEQ" spanning from 14:6 to 14:10

IR syntax error: Can't recover from previous error(s)

Content of test case:

pa/pa4/grading/tests/base/assign06.xi

```
1 use io
2 use conv
3
4 main(args: int[][] ) {
5   a: int[3]
6   a[f(0)] = f(3)
7   a[f(1)] = f(4)
8   a[f(2)] = f(5)
9   println(unparseInt(length(a)))
10  println(unparseInt(a[1]))
11 }
12
13 f(x: int): int {
14   print("f() called: ")
15   println(unparseInt(x))
16   return x
17 }
```

Generated result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/assign06.ir.nml

Expected result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/assign06.irsol.nml

```
1 f() called: 0
2 f() called: 3
3 f() called: 1
4 f() called: 4
5 f() called: 2
6 f() called: 5
7 3
8 4
```

Generated result for --irgen:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/assign06.ir

```
1 (COMPUNIT
2   assign06
3   (FUNC
4     _Imain_pai
5     (SEQ
6       (MOVE (TEMP args_main) (TEMP _ARG0))
7       (MOVE (TEMP _LEN_0) (CONST 3))
8       (MOVE (TEMP _ARR_0)
9         (CALL (NAME _I_alloc_i) (MUL (ADD (TEMP _LEN_0) (CONST 1)) (CONST 8))))
10      (MOVE (MEM (TEMP _ARR_0)) (TEMP _LEN_0))
11      (MOVE (TEMP _ARR_0) (ADD (TEMP _ARR_0) (CONST 8)))
12      (MOVE (TEMP a_main) (TEMP _ARR_0))
13      (MOVE
14        (ESEQ
15          (SEQ
16            (MOVE (TEMP _var_11) (TEMP a_main))
17            (MOVE (TEMP __var_1) (CALL (NAME _If_ii) (CONST 0))))))
```

```

18      (MEM (ADD (TEMP _var_11) (MUL (CONST 8) (TEMP __var_1))))))
19      (CALL (NAME _If_ii) (CONST 3)))
20  (MOVE
21      (ESEQ
22          (SEQ
23              (MOVE (TEMP _var_12) (TEMP a_main))
24              (MOVE (TEMP __var_2) (CALL (NAME _If_ii) (CONST 1))))
25              (MEM (ADD (TEMP _var_12) (MUL (CONST 8) (TEMP __var_2))))))
26          (CALL (NAME _If_ii) (CONST 4)))
27  (MOVE
28      (ESEQ
29          (SEQ
30              (MOVE (TEMP _var_13) (TEMP a_main))
31              (MOVE (TEMP __var_3) (CALL (NAME _If_ii) (CONST 2))))
32              (MEM (ADD (TEMP _var_13) (MUL (CONST 8) (TEMP __var_3))))))
33          (CALL (NAME _If_ii) (CONST 5)))
34  (MOVE (TEMP __var_5)
35      (CALL (NAME _IunparseInt_aii) (MEM (SUB (TEMP a_main) (CONST 8)))))
36  (MOVE (TEMP __var_4) (CALL (NAME _Iprintln_pai) (TEMP __var_5)))
37  (MOVE (TEMP __var_7)
38      (CALL (NAME _IunparseInt_aii) (MEM (ADD (TEMP a_main) (CONST 8)))))
39  (MOVE (TEMP __var_6) (CALL (NAME _Iprintln_pai) (TEMP __var_7)))
40  (RETURN)))
41  (FUNC
42      _If_ii
43      (SEQ
44          (MOVE (TEMP x_f) (TEMP _ARG0))
45          (MOVE (TEMP _ARR_1) (CALL (NAME _I_alloc_i) (CONST 104)))
46          (MOVE (MEM (TEMP _ARR_1)) (CONST 12))
47          (MOVE (TEMP _ARR_1) (ADD (TEMP _ARR_1) (CONST 8)))
48          (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 0))) (CONST 102))
49          (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 8))) (CONST 40))
50          (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 16))) (CONST 41))
51          (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 24))) (CONST 32))
52          (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 32))) (CONST 99))
53          (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 40))) (CONST 97))
54          (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 48))) (CONST 108))
55          (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 56))) (CONST 108))
56          (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 64))) (CONST 101))
57          (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 72))) (CONST 100))
58          (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 80))) (CONST 58))
59          (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 88))) (CONST 32))
60          (MOVE (TEMP __var_8) (CALL (NAME _Iprint_pai) (TEMP _ARR_1)))
61          (MOVE (TEMP __var_10) (CALL (NAME _IunparseInt_aii) (TEMP x_f)))
62          (MOVE (TEMP __var_9) (CALL (NAME _Iprintln_pai) (TEMP __var_10)))
63          (MOVE (TEMP _RET0) (TEMP x_f))
64          (RETURN)
65          (RETURN))))
66

```

xic-ref (--irgen [basic test]): assign07.xi

Syntax error for input symbol "ESEQ" spanning from 17:6 to 17:10

instead expected token classes are []

Couldn't repair and continue parse for input symbol "ESEQ" spanning from 17:6 to 17:10

IR syntax error: Can't recover from previous error(s)

Content of test case:

pa/pa4/grading/tests/base/assign07.xi

```
1 use io
2 use conv
3
4 main(args: int[][]) {
5   a: int[] = {1, 2, 3}
6   a[a[f(a, 1)]] = a[g(a, 2)]
7   println(unparseInt(a[0]))
8   println(unparseInt(a[1]))
9   println(unparseInt(a[2]))
10 }
11
12 f(a: int[], i: int): int {
13   print("f() called: ")
14   println(unparseInt(i))
15   return i
16 }
17
18 g(a: int[], i: int): int {
19   print("g() called: ")
20   println(unparseInt(i))
21   a[i] = 47
22   return i
23 }
```

Generated result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/assign07.ir.nml

Expected result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/assign07.irsol.nml

```
1 f() called: 1
2 g() called: 2
3 1
4 2
5 47
```

Generated result for --irgen:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/assign07.ir

```
1 (COMPUNIT
2   assign07
3   (FUNC
4     _Imain_paii
5     (SEQ
6       (MOVE (TEMP args_main) (TEMP _ARG0))
7       (MOVE (TEMP _ARR_0) (CALL (NAME _I_alloc_i) (CONST 32)))
8       (MOVE (MEM (TEMP _ARR_0)) (CONST 3))
9       (MOVE (TEMP _ARR_0) (ADD (TEMP _ARR_0) (CONST 8)))
10      (MOVE (MEM (ADD (TEMP _ARR_0) (CONST 0))) (CONST 1))
11      (MOVE (MEM (ADD (TEMP _ARR_0) (CONST 8))) (CONST 2))
12      (MOVE (MEM (ADD (TEMP _ARR_0) (CONST 16))) (CONST 3))
13      (MOVE (TEMP a_main) (TEMP _ARR_0))
14      (MOVE (TEMP _var_15) (TEMP a_main))
```

```

15  (MOVE (TEMP __var_1) (CALL (NAME _Ig_iaii) (TEMP a_main) (CONST 2)))
16  (MOVE
17  (ESEQ
18  (SEQ
19  (SEQ
20  (SEQ
21  (MOVE (TEMP _var_17) (TEMP a_main))
22  (MOVE (TEMP _var_14) (TEMP a_main)))
23  (SEQ
24  (MOVE (TEMP _var_18) (TEMP _var_17))
25  (MOVE (TEMP _var_16) (CONST 8))))
26  (SEQ
27  (MOVE (TEMP _var_19) (TEMP _var_18))
28  (MOVE (TEMP __var_0) (CALL (NAME _If_iaii) (TEMP a_main) (CONST 1))))))
29  (MEM
30  (ADD
31  (TEMP _var_19)
32  (MUL (TEMP _var_16)
33  (MEM (ADD (TEMP _var_14) (MUL (CONST 8) (TEMP __var_0)))))))
34  (MEM (ADD (TEMP _var_15) (MUL (CONST 8) (TEMP __var_1))))
35  (MOVE (TEMP __var_3)
36  (CALL (NAME _IunparseInt_aii) (MEM (ADD (TEMP a_main) (CONST 0))))))
37  (MOVE (TEMP __var_2) (CALL (NAME _Iprintln_pai) (TEMP __var_3)))
38  (MOVE (TEMP __var_5)
39  (CALL (NAME _IunparseInt_aii) (MEM (ADD (TEMP a_main) (CONST 8))))))
40  (MOVE (TEMP __var_4) (CALL (NAME _Iprintln_pai) (TEMP __var_5)))
41  (MOVE (TEMP __var_7)
42  (CALL (NAME _IunparseInt_aii) (MEM (ADD (TEMP a_main) (CONST 16))))))
43  (MOVE (TEMP __var_6) (CALL (NAME _Iprintln_pai) (TEMP __var_7)))
44  (RETURN)))
45  (FUNC
46  _If_iaii
47  (SEQ
48  (MOVE (TEMP a_f) (TEMP _ARG0))
49  (MOVE (TEMP i_f) (TEMP _ARG1))
50  (MOVE (TEMP _ARR_1) (CALL (NAME _I_alloc_i) (CONST 104)))
51  (MOVE (MEM (TEMP _ARR_1)) (CONST 12))
52  (MOVE (TEMP _ARR_1) (ADD (TEMP _ARR_1) (CONST 8)))
53  (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 0))) (CONST 102))
54  (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 8))) (CONST 40))
55  (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 16))) (CONST 41))
56  (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 24))) (CONST 32))
57  (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 32))) (CONST 99))
58  (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 40))) (CONST 97))
59  (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 48))) (CONST 108))
60  (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 56))) (CONST 108))
61  (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 64))) (CONST 101))
62  (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 72))) (CONST 100))
63  (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 80))) (CONST 58))
64  (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 88))) (CONST 32))
65  (MOVE (TEMP __var_8) (CALL (NAME _Iprint_pai) (TEMP _ARR_1)))
66  (MOVE (TEMP __var_10) (CALL (NAME _IunparseInt_aii) (TEMP i_f)))
67  (MOVE (TEMP __var_9) (CALL (NAME _Iprintln_pai) (TEMP __var_10)))
68  (MOVE (TEMP _RET0) (TEMP i_f))
69  (RETURN)
70  (RETURN)))
71  (FUNC
72  _Ig_iaii
73  (SEQ
74  (MOVE (TEMP a_g) (TEMP _ARG0))

```

```

75  (MOVE (TEMP i_g) (TEMP _ARG1))
76  (MOVE (TEMP _ARR_2) (CALL (NAME _I_alloc_i) (CONST 104)))
77  (MOVE (MEM (TEMP _ARR_2)) (CONST 12))
78  (MOVE (TEMP _ARR_2) (ADD (TEMP _ARR_2) (CONST 8)))
79  (MOVE (MEM (ADD (TEMP _ARR_2) (CONST 0))) (CONST 103))
80  (MOVE (MEM (ADD (TEMP _ARR_2) (CONST 8))) (CONST 40))
81  (MOVE (MEM (ADD (TEMP _ARR_2) (CONST 16))) (CONST 41))
82  (MOVE (MEM (ADD (TEMP _ARR_2) (CONST 24))) (CONST 32))
83  (MOVE (MEM (ADD (TEMP _ARR_2) (CONST 32))) (CONST 99))
84  (MOVE (MEM (ADD (TEMP _ARR_2) (CONST 40))) (CONST 97))
85  (MOVE (MEM (ADD (TEMP _ARR_2) (CONST 48))) (CONST 108))
86  (MOVE (MEM (ADD (TEMP _ARR_2) (CONST 56))) (CONST 108))
87  (MOVE (MEM (ADD (TEMP _ARR_2) (CONST 64))) (CONST 101))
88  (MOVE (MEM (ADD (TEMP _ARR_2) (CONST 72))) (CONST 100))
89  (MOVE (MEM (ADD (TEMP _ARR_2) (CONST 80))) (CONST 58))
90  (MOVE (MEM (ADD (TEMP _ARR_2) (CONST 88))) (CONST 32))
91  (MOVE (TEMP __var_11) (CALL (NAME _Iprint_pai) (TEMP _ARR_2)))
92  (MOVE (TEMP __var_13) (CALL (NAME _IunparseInt_aii) (TEMP i_g)))
93  (MOVE (TEMP __var_12) (CALL (NAME _Iprintln_pai) (TEMP __var_13)))
94  (MOVE (MEM (ADD (TEMP a_g) (MUL (TEMP i_g) (CONST 8)))) (CONST 47))
95  (MOVE (TEMP _RET0) (TEMP i_g))
96  (RETURN)
97  (RETURN)))
98

```

xic-ref (--irgen [basic test]): assign08.xi

Syntax error for input symbol "ESEQ" spanning from 17:6 to 17:10

instead expected token classes are []

Couldn't repair and continue parse for input symbol "ESEQ" spanning from 17:6 to 17:10

IR syntax error: Can't recover from previous error(s)

Content of test case:

pa/pa4/grading/tests/base/assign08.xi

```
1 use io
2 use conv
3
4 main(args: int[][]) {
5   a: int[] = {1, 2, 3}
6   a[f(a, 1)] = a[g(a, 2)]
7 }
8
9 f(a: int[], i: int): int {
10  print("f() called: ")
11  println(unparseInt(i))
12  a[i] = 25
13  return i
14 }
15
16 g(a: int[], i: int): int {
17  print("g() called: ")
18  println(unparseInt(i))
19  a[i] = 47
20  return i
21 }
```

Generated result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/assign08.ir.nml

Expected result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/assign08.irsol.nml

1 f() called: 1

2 g() called: 2

Generated result for --irgen:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/assign08.ir

```
1 (COMPUNIT
2  assign08
3  (FUNC
4    _Imain_paii
5    (SEQ
6      (MOVE (TEMP args_main) (TEMP _ARG0))
7      (MOVE (TEMP _ARR_0) (CALL (NAME _I_alloc_i) (CONST 32)))
8      (MOVE (MEM (TEMP _ARR_0)) (CONST 3))
9      (MOVE (TEMP _ARR_0) (ADD (TEMP _ARR_0) (CONST 8)))
10     (MOVE (MEM (ADD (TEMP _ARR_0) (CONST 0))) (CONST 1))
11     (MOVE (MEM (ADD (TEMP _ARR_0) (CONST 8))) (CONST 2))
12     (MOVE (MEM (ADD (TEMP _ARR_0) (CONST 16))) (CONST 3))
13     (MOVE (TEMP a_main) (TEMP _ARR_0))
14     (MOVE (TEMP _var_9) (TEMP a_main))
15     (MOVE (TEMP __var_1) (CALL (NAME _Ig_iai) (TEMP a_main) (CONST 2)))
16     (MOVE
17       (ESEQ
18         (SEQ
19           (MOVE (TEMP _var_8) (TEMP a_main))
```

```

20      (MOVE (TEMP __var_0) (CALL (NAME _If_iai) (TEMP a_main) (CONST 1)))
21      (MEM (ADD (TEMP _var_8) (MUL (CONST 8) (TEMP __var_0))))
22      (MEM (ADD (TEMP _var_9) (MUL (CONST 8) (TEMP __var_1))))
23      (RETURN)))
24 (FUNC
25   _If_iai
26   (SEQ
27    (MOVE (TEMP a_f) (TEMP _ARG0))
28    (MOVE (TEMP i_f) (TEMP _ARG1))
29    (MOVE (TEMP _ARR_1) (CALL (NAME _I_alloc_i) (CONST 104)))
30    (MOVE (MEM (TEMP _ARR_1)) (CONST 12))
31    (MOVE (TEMP _ARR_1) (ADD (TEMP _ARR_1) (CONST 8)))
32    (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 0))) (CONST 102))
33    (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 8))) (CONST 40))
34    (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 16))) (CONST 41))
35    (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 24))) (CONST 32))
36    (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 32))) (CONST 99))
37    (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 40))) (CONST 97))
38    (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 48))) (CONST 108))
39    (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 56))) (CONST 108))
40    (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 64))) (CONST 101))
41    (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 72))) (CONST 100))
42    (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 80))) (CONST 58))
43    (MOVE (MEM (ADD (TEMP _ARR_1) (CONST 88))) (CONST 32))
44    (MOVE (TEMP __var_2) (CALL (NAME _Iprint_pai) (TEMP _ARR_1)))
45    (MOVE (TEMP __var_4) (CALL (NAME _IunparseInt_iai) (TEMP i_f)))
46    (MOVE (TEMP __var_3) (CALL (NAME _Iprintln_pai) (TEMP __var_4)))
47    (MOVE (MEM (ADD (TEMP a_f) (MUL (TEMP i_f) (CONST 8)))) (CONST 25))
48    (MOVE (TEMP _RET0) (TEMP i_f))
49    (RETURN)
50    (RETURN)))
51 (FUNC
52   _Ig_iai
53   (SEQ
54    (MOVE (TEMP a_g) (TEMP _ARG0))
55    (MOVE (TEMP i_g) (TEMP _ARG1))
56    (MOVE (TEMP _ARR_2) (CALL (NAME _I_alloc_i) (CONST 104)))
57    (MOVE (MEM (TEMP _ARR_2)) (CONST 12))
58    (MOVE (TEMP _ARR_2) (ADD (TEMP _ARR_2) (CONST 8)))
59    (MOVE (MEM (ADD (TEMP _ARR_2) (CONST 0))) (CONST 103))
60    (MOVE (MEM (ADD (TEMP _ARR_2) (CONST 8))) (CONST 40))
61    (MOVE (MEM (ADD (TEMP _ARR_2) (CONST 16))) (CONST 41))
62    (MOVE (MEM (ADD (TEMP _ARR_2) (CONST 24))) (CONST 32))
63    (MOVE (MEM (ADD (TEMP _ARR_2) (CONST 32))) (CONST 99))
64    (MOVE (MEM (ADD (TEMP _ARR_2) (CONST 40))) (CONST 97))
65    (MOVE (MEM (ADD (TEMP _ARR_2) (CONST 48))) (CONST 108))
66    (MOVE (MEM (ADD (TEMP _ARR_2) (CONST 56))) (CONST 108))
67    (MOVE (MEM (ADD (TEMP _ARR_2) (CONST 64))) (CONST 101))
68    (MOVE (MEM (ADD (TEMP _ARR_2) (CONST 72))) (CONST 100))
69    (MOVE (MEM (ADD (TEMP _ARR_2) (CONST 80))) (CONST 58))
70    (MOVE (MEM (ADD (TEMP _ARR_2) (CONST 88))) (CONST 32))
71    (MOVE (TEMP __var_5) (CALL (NAME _Iprint_pai) (TEMP _ARR_2)))
72    (MOVE (TEMP __var_7) (CALL (NAME _IunparseInt_iai) (TEMP i_g)))
73    (MOVE (TEMP __var_6) (CALL (NAME _Iprintln_pai) (TEMP __var_7)))
74    (MOVE (MEM (ADD (TEMP a_g) (MUL (TEMP i_g) (CONST 8)))) (CONST 47))
75    (MOVE (TEMP _RET0) (TEMP i_g))
76    (RETURN)
77    (RETURN)))
78

```

xic-ref (--irgen [basic test]): assign09.xi

Syntax error for input symbol "NUMBER" spanning from 1:1 to 1:3

instead expected token classes are []

Couldn't repair and continue parse for input symbol "NUMBER" spanning from 1:1 to 1:3

IR syntax error: Can't recover from previous error(s)

Content of test case:

pa/pa4/grading/tests/base/assign09.xi

```
1 use io
2 use conv
3
4 main(args: int[][]) {
5   x: int[] = {1, 2, 3, 4}
6   a: int[5][4]
7   {
8     i: int = 0
9     while (i < length(a)) {
10      j: int = 0
11      while (j < length(a[i])) {
12        a[i][j] = i + j
13        j = j + 1
14      }
15      i = i + 1
16    }
17  }
18  a[f(x,3)-f(x,2)][f(x,2)-f(x,1)] = 47
19  {
20    i: int = 0
21    while (i < length(a)) {
22      j: int = 0
23      while (j < length(a[i])) {
24        println(unparseInt(a[i][j]))
25        j = j + 1
26      }
27      i = i + 1
28    }
29  }
30 }
31
32 f(a: int[], i: int): int {
33   print("Index: ")
34   println(unparseInt(i))
35   a[i] = a[i] + 1
36   return a[i]
37 }
```

Compiler's standard output:

Syntax error at /home/vagrant/shared/xth-

workpath/Submissions/group_of_fl356_rw489_vs442_zr54/assign09.xi: 11:26: Unexpected token [

Compiler's standard error:

null

Generated result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/assign09.ir.nml

Expected result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/assign09.irsol.nml

1 Index: 3

xic-ref (--irgen [basic test]): assign09.xi

```
2 Index: 2
3 Index: 2
4 Index: 1
5 0
6 1
7 2
8 3
9 1
10 2
11 47
12 4
13 2
14 3
15 4
16 5
17 3
18 4
19 5
20 6
21 4
22 5
23 6
24 7
```

Generated result for --irgen:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/assign09.ir

```
1 11:26 error:Unexpected token [
```

```
2
```

xic-ref (--irgen [basic test]): assign10.xi

Syntax error for input symbol "NUMBER" spanning from 1:1 to 1:3

instead expected token classes are []

Couldn't repair and continue parse for input symbol "NUMBER" spanning from 1:1 to 1:3

IR syntax error: Can't recover from previous error(s)

Content of test case:

pa/pa4/grading/tests/base/assign10.xi

```
1 use io
2 use conv
3
4 main(args: int[][] ) {
5   a: int[5][4]
6   {
7     i: int = 0
8     while (i < length(a)) {
9       j: int = 0
10      while (j < length(a[i])) {
11        a[i][j] = i + j
12        j = j + 1
13      }
14      i = i + 1
15    }
16  }
17  a[f(a, 4, 2)-3][f(a, 4, 2)-6] = 47
18  {
19    i: int = 0
20    while (i < length(a)) {
21      j: int = 0
22      while (j < length(a[i])) {
23        println(unparseInt(a[i][j]))
24        j = j + 1
25      }
26      i = i + 1
27    }
28  }
29 }
30
31 f(a: int[][] , i: int, j: int): int {
32   print("Index: ")
33   print(unparseInt(i))
34   print(" ")
35   println(unparseInt(j))
36   a[i][j] = a[i][j] + 1
37   return a[i][j]
38 }
```

Compiler's standard output:

Syntax error at /home/vagrant/shared/xth-

workpath/Submissions/group_of_fl356_rw489_vs442_zr54/assign10.xi: 10:26: Unexpected token [

Compiler's standard error:

null

Generated result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/assign10.ir.nml

Expected result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/assign10.irsol.nml

xic-ref (--irgen [basic test]): assign10.xi

```
1 Index: 4 2
2 Index: 4 2
3 0
4 1
5 2
6 3
7 1
8 2
9 3
10 4
11 2
12 3
13 4
14 5
15 3
16 4
17 5
18 6
19 4
20 5
21 47
22 7
```

Generated result for --irgen:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/assign10.ir

```
1 10:26 error:Unexpected token [
```

```
2
```

xic-ref (--irgen [basic test]): binary09.xi

Syntax error for input symbol "EOF" spanning from 3:1 to 3:1

instead expected token classes are []

Couldn't repair and continue parse for input symbol "EOF" spanning from 3:1 to 3:1

IR syntax error: Can't recover from previous error(s)

Content of test case:

pa/pa4/grading/tests/base/binary09.xi

```
1 use io
2 use conv
3
4 main(args: int[][]) {
5   a1: int[] = {1, 2, 3, 4}
6   a2: int[] = {5, 6, 7}
7   a3: int[] = {}
8   printai(a1 + a2)
9   printai(a1 + a3)
10  printai(a2 + a3)
11 }
12
13 printai(a: int[]) {
14   i: int = 0
15   print("{")
16   while (i < length(a)) {
17     if (i > 0) print(", ")
18     print(unparseInt(a[i]))
19     i = i + 1
20   }
21   println("}")
22 }
```

Compiler's standard error:

Index: 0, Size: 0

Generated result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/binary09.ir.nml

Expected result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/binary09.irsol.nml

```
1 {1, 2, 3, 4, 5, 6, 7}
2 {1, 2, 3, 4}
3 {5, 6, 7}
```

Generated result for --irgen:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/binary09.ir

```
1
2
```

xic-ref (--irgen [basic test]): binary10.xi

Syntax error for input symbol "EOF" spanning from 3:1 to 3:1

instead expected token classes are []

Couldn't repair and continue parse for input symbol "EOF" spanning from 3:1 to 3:1

IR syntax error: Can't recover from previous error(s)

Content of test case:

pa/pa4/grading/tests/base/binary10.xi

```
1 use io
2 use conv
3
4 main(args: int[][]) {
5   a1: int[] = {1, 2, 3, 4}
6   a2: int[] = {5, 6, 7}
7   a3: int[] = {}
8   printai(f(a1) + f(a2))
9   printai(f(a1) + f(a3))
10  printai(f(a2) + f(a3))
11 }
12
13 f(x: int[]): int[] {
14   print("f() called: ")
15   printai(x)
16   return x
17 }
18
19 printai(a: int[]) {
20   i: int = 0
21   print("{}")
22   while (i < length(a)) {
23     if (i > 0) print(", ")
24     print(unparseInt(a[i]))
25     i = i + 1
26   }
27   println("{}")
28 }
```

Compiler's standard error:

Index: 0, Size: 0

Generated result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/binary10.ir.nml

Expected result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/binary10.irsol.nml

```
1 f() called: {1, 2, 3, 4}
2 f() called: {5, 6, 7}
3 {1, 2, 3, 4, 5, 6, 7}
4 f() called: {1, 2, 3, 4}
5 f() called: {}
6 {1, 2, 3, 4}
7 f() called: {5, 6, 7}
8 f() called: {}
9 {5, 6, 7}
```

Generated result for --irgen:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/binary10.ir

```
1
2
```


xic-ref (--irgen [basic test]): length02.xi

Syntax error for input symbol "NUMBER" spanning from 1:1 to 1:3
instead expected token classes are []
Couldn't repair and continue parse for input symbol "NUMBER" spanning from 1:1 to 1:3

IR syntax error: Can't recover from previous error(s)

Content of test case:

pa/pa4/grading/tests/base/length02.xi

```
1 use io
2 use conv
3
4 main(args: int[][]) {
5   a: int[][] = {{1}, {2, 3}, {4, 5, 6}, {7, 8, 9, 10}}
6   println(unparseInt(length(a)))
7   {
8     i: int = 0
9     while (i < length(a)) {
10      println(unparseInt(length(a[i])))
11      i = i + 1
12    }
13  }
14 }
```

Compiler's standard output:

Syntax error at /home/vagrant/shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/length02.xi: 10:34: Unexpected token [

Compiler's standard error:

null

Generated result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/length02.ir.nml

Expected result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/length02.irsol.nml

```
1 4
2 1
3 2
4 3
5 4
```

Generated result for --irgen:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/length02.ir

```
1 10:34 error:Unexpected token [
2
```

xic-ref (--irgen [basic test]): localdecl06.xi

Syntax error for input symbol "NUMBER" spanning from 1:1 to 1:3
instead expected token classes are []
Couldn't repair and continue parse for input symbol "NUMBER" spanning from 1:1 to 1:3

IR syntax error: Can't recover from previous error(s)

Content of test case:

pa/pa4/grading/tests/base/localdecl06.xi

```
1 use io
2 use conv
3
4 main(args: int[][]){
5   a1: int[25][][][][]
6   a2: int[47][][][]
7   a3: int[20][][]
8   a4: int[16][]
9   a5: int[42]
10  a1[17] = a2
11  a1[17][16] = a3
12  a1[17][16][15] = a4
13  a1[17][16][15][14] = a5
14  a1[17][16][15][14][13] = 12
15  println(unparseInt(length(a1)))
16  println(unparseInt(length(a1[17])))
17  println(unparseInt(length(a1[17][16])))
18  println(unparseInt(length(a1[17][16][15])))
19  println(unparseInt(length(a1[17][16][15][14])))
20  println(unparseInt(a1[17][16][15][14][13]))
21  println("done")
22 }
```

Compiler's standard output:

Syntax error at /home/vagrant/shared/xth-
workpath/Submissions/group_of_fl356_rw489_vs442_zr54/localdecl06.xi: 16:31: Unexpected token [

Compiler's standard error:

null

Generated result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/localdecl06.ir.nml

Expected result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/localdecl06.irsol.nml

```
1 25
2 47
3 20
4 16
5 42
6 12
7 done
```

Generated result for --irgen:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/localdecl06.ir

```
1 16:31 error:Unexpected token [
2
```

xic-ref (--irgen [basic test]): localdecl08.xi

Syntax error for input symbol "EOF" spanning from 3:1 to 3:1

instead expected token classes are []

Couldn't repair and continue parse for input symbol "EOF" spanning from 3:1 to 3:1

IR syntax error: Can't recover from previous error(s)

Content of test case:

pa/pa4/grading/tests/base/localdecl08.xi

```
1 use io
2 use conv
3
4 main(args: int[][] ) {
5   _ = f()
6   println("done")
7 }
8
9 f(): int {
10  println("f() called")
11  return 17
12 }
```

Compiler's standard error:

null

Generated result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/localdecl08.ir.nml

Expected result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/localdecl08.irsol.nml

```
1 f() called
2 done
```

Generated result for --irgen:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/localdecl08.ir

```
1
2
```

xic-ref (--irgen [basic test]): localdecl10.xi

Syntax error for input symbol "EOF" spanning from 3:1 to 3:1

instead expected token classes are []

Couldn't repair and continue parse for input symbol "EOF" spanning from 3:1 to 3:1

IR syntax error: Can't recover from previous error(s)

Content of test case:

pa/pa4/grading/tests/base/localdecl10.xi

```
1 use io
2 use conv
3
4 main(args: int[][] ) {
5   x:int, _, z: int = f()
6   println(unparseInt(x))
7   println(unparseInt(z))
8   println("done")
9 }
10
11 f(): int, bool, int {
12   println("f() called")
13   return 17, true, 42
14 }
```

Compiler's standard error:

null

Generated result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/localdecl10.ir.nml

Expected result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/localdecl10.irsol.nml

```
1 f() called
2 17
3 42
4 done
```

Generated result for --irgen:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/localdecl10.ir

```
1
2
```

xic-ref (--irgen [basic test]): localdecl11.xi

Syntax error for input symbol "EOF" spanning from 3:1 to 3:1

instead expected token classes are []

Couldn't repair and continue parse for input symbol "EOF" spanning from 3:1 to 3:1

IR syntax error: Can't recover from previous error(s)

Content of test case:

pa/pa4/grading/tests/base/localdecl11.xi

```
1 use io
2 use conv
3
4 main(args: int[][] ) {
5   _, y: bool, z: int = f()
6   if (y) println("true") else println("false")
7   println(unparseInt(z))
8   println("done")
9 }
10
11 f(): int, bool, int {
12   println("f() called")
13   return 17, true, 42
14 }
```

Compiler's standard error:

null

Generated result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/localdecl11.ir.nml

Expected result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/localdecl11.irsol.nml

```
1 f() called
2 true
3 42
4 done
```

Generated result for --irgen:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/localdecl11.ir

```
1
2
```

xic-ref (--irgen [basic test]): localdecl12.xi

Syntax error for input symbol "EOF" spanning from 3:1 to 3:1

instead expected token classes are []

Couldn't repair and continue parse for input symbol "EOF" spanning from 3:1 to 3:1

IR syntax error: Can't recover from previous error(s)

Content of test case:

pa/pa4/grading/tests/base/localdecl12.xi

```
1 use io
2 use conv
3
4 main(args: int[][]) {
5   _, _, _ = f()
6   println("done")
7 }
8
9 f(): int, bool, int {
10  println("f() called")
11  return 17, true, 42
12 }
```

Compiler's standard error:

null

Generated result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/localdecl12.ir.nml

Expected result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/localdecl12.irsol.nml

```
1 f() called
2 done
```

Generated result for --irgen:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/localdecl12.ir

```
1
2
```

xic-ref (--irgen [basic test]): localdecl13.xi

Syntax error for input symbol "NUMBER" spanning from 1:1 to 1:2
instead expected token classes are []
Couldn't repair and continue parse for input symbol "NUMBER" spanning from 1:1 to 1:2

IR syntax error: Can't recover from previous error(s)

Content of test case:

pa/pa4/grading/tests/base/localdecl13.xi

```
1 use io
2 use conv
3
4 main(args: int[][]) {
5   a: int[] = {1, 2, 3}
6   x: int[f(a, 0)][f(a, 0)][f(a, 0)][][]
7   println(unparseInt(length(x)))
8   println(unparseInt(length(x[0])))
9   println(unparseInt(length(x[0][0])))
10  println(unparseInt(a[0]))
11  println(unparseInt(a[1]))
12  println(unparseInt(a[2]))
13 }
14
15 f(a: int[], i: int): int {
16   print("Index: ")
17   println(unparseInt(i))
18   a[i] = a[i] + 1
19   return a[i]
20 }
```

Compiler's standard output:

Syntax error at /home/vagrant/shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/localdecl13.xi: 8:30: Unexpected token [

Compiler's standard error:

null

Generated result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/localdecl13.ir.nml

Expected result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/localdecl13.irsol.nml

```
1 Index: 0
2 Index: 0
3 Index: 0
4 2
5 3
6 4
7 4
8 2
9 3
```

Generated result for --irgen:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/localdecl13.ir

```
1 8:30 error:Unexpected token [
2
```

xic-ref (--irgen [basic test]): localdecl14.xi

Syntax error for input symbol "NUMBER" spanning from 1:1 to 1:2

instead expected token classes are []

Couldn't repair and continue parse for input symbol "NUMBER" spanning from 1:1 to 1:2

IR syntax error: Can't recover from previous error(s)

Content of test case:

pa/pa4/grading/tests/base/localdecl14.xi

```
1 use io
2 use conv
3
4 main(args: int[][]) {
5   a: int[] = {1, 2, 3, 4}
6   x: int[f(a,3)-f(a,2)][f(a,2)-f(a,1)][f(a,1)-f(a,0)][[]]
7   println(unparseInt(length(x)))
8   println(unparseInt(length(x[0])))
9   println(unparseInt(length(x[0][0])))
10  println(unparseInt(a[0]))
11  println(unparseInt(a[1]))
12  println(unparseInt(a[2]))
13  println(unparseInt(a[3]))
14 }
15
16 f(a: int[], i: int): int {
17   print("Index: ")
18   println(unparseInt(i))
19   a[i] = a[i] + 1
20   return a[i]
21 }
```

Compiler's standard output:

Syntax error at /home/vagrant/shared/xth-

workpath/Submissions/group_of_fl356_rw489_vs442_zr54/localdecl14.xi: 8:30: Unexpected token [

Compiler's standard error:

null

Generated result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/localdecl14.ir.nml

Expected result for --irrun:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/localdecl14.irsol.nml

```
1 Index: 3
2 Index: 2
3 Index: 2
4 Index: 1
5 Index: 1
6 Index: 0
7 1
8 2
9 2
10 2
11 4
12 5
13 5
```

Generated result for --irgen:

shared/xth-workpath/Submissions/group_of_fl356_rw489_vs442_zr54/localdecl14.ir

1 8:30 error:Unexpected token [

xic-ref (--irgen [basic test]): localdecl14.xi

2